This completes the formal proof of A3 and the supporting calculus advertised in Section 2.

Appendix B. Irrep Projectors & No-Go for $q_{\lambda\mu\nu}(v)$

This appendix collects the group-theoretic ingredients used in Section 4: (i) the irreducible decomposition of the torsion tensor under the Lorentz group, (ii) explicit, idempotent projectors onto the trace, axial, and traceless sectors, (iii) the quadratic identity for $T^A{}_{BC}T_A{}^{BC}$ in our conventions, and (iv) the *single-vector no-go* (Proposition A3) for the $q_{\lambda\mu\nu}$ irrep.

All statements are purely algebraic and hold before/after applying the scalar projector $\Pi_{\mathrm{PT}}[\cdot]$; after projection, all scalar contractions are real (Section 2).

Appendix B.1. Torsion as a Lorentz Representation and Its Algebra

In index language (spacetime indices), torsion is a rank-3 tensor antisymmetric in its last two indices, $T_{\lambda\mu\nu} = -T_{\lambda\nu\mu}$, with $4 \times 6 = 24$ independent components in $d = 4$. The Lorentz-covariant irreducible content splits into

$$\mathbf{24} \simeq \underbrace{\mathbf{4}}_{\text{trace } T_\mu} \oplus \underbrace{\mathbf{4}}_{\text{axial } S^\mu} \oplus \underbrace{\mathbf{16}}_{\text{traceless tensor } q_{\lambda\mu\nu}}, \qquad \begin{cases} T_\mu \equiv T^\nu{}_{\mu\nu}, \\ S^\mu \equiv \epsilon^{\mu\nu\rho\sigma} T_{\nu\rho\sigma}, \\ q^\nu{}_{\mu\nu} = q_{\lambda\mu}{}^\mu = \epsilon^{\mu\nu\rho\sigma} q_{\nu\rho\sigma} = 0. \end{cases} \tag{A13}$$

We use $\epsilon^{0123} = +1$ and the metric signature $(-,+,+,+)$, and we adopt the standard scalar product $(X, Y) \equiv X_{\lambda\mu\nu} Y^{\lambda\mu\nu}$ on this space. (Frame and spacetime contractions are equivalent once $e^A{}_\mu$ is inserted; all formulas below hold with either $T^A{}_{BC}$ or $T_{\lambda\mu\nu}$, and we freely move between the two notations.)

Appendix B.2. Idempotent Projectors

Define three linear maps $P^{(T)}, P^{(S)}, P^{(q)}$ on the torsion space by

$$\left(P^{(T)}T\right)_{\lambda\mu\nu} \equiv \frac{1}{3}\left(g_{\lambda\mu}T_\nu - g_{\lambda\nu}T_\mu\right), \tag{A14}$$

$$\left(P^{(S)}T\right)_{\lambda\mu\nu} \equiv -\frac{1}{6}\,\epsilon_{\lambda\mu\nu\rho}\,S^\rho \;=\; -\frac{1}{6}\,\epsilon_{\lambda\mu\nu\rho}\,\epsilon^{\rho\alpha\beta\gamma}T_{\alpha\beta\gamma}, \tag{A15}$$

$$\left(P^{(q)}T\right)_{\lambda\mu\nu} \equiv T_{\lambda\mu\nu} - \left(P^{(T)}T\right)_{\lambda\mu\nu} - \left(P^{(S)}T\right)_{\lambda\mu\nu}. \tag{A16}$$

These are the unique Lorentz-covariant, algebraic (derivative-free) projectors onto the three irreps in Equation (A13). A direct computation shows

$$\left(P^{(X)}\right)^2 = P^{(X)}, \qquad P^{(X)}P^{(Y)} = 0 \;(X \neq Y), \qquad P^{(T)} + P^{(S)} + P^{(q)} = \mathbf{1}, \tag{A17}$$

and the images obey by construction the following relations:

$$\begin{aligned} \left(P^{(T)}T\right)^\nu{}_{\mu\nu} &= T_\mu, & \epsilon^{\mu\nu\rho\sigma}\left(P^{(T)}T\right)_{\nu\rho\sigma} &= 0, \\ \left(P^{(S)}T\right)^\nu{}_{\mu\nu} &= 0, & \epsilon^{\mu\nu\rho\sigma}\left(P^{(S)}T\right)_{\nu\rho\sigma} &= S^\mu, \\ \left(P^{(q)}T\right)^\nu{}_{\mu\nu} &= 0, & \epsilon^{\mu\nu\rho\sigma}\left(P^{(q)}T\right)_{\nu\rho\sigma} &= 0. \end{aligned} \tag{A18}$$

Orthogonality and quadratic split.

With the scalar product $(\cdot,\cdot)$,

$$\left(P^{(X)}T,\, P^{(Y)}T\right) = 0 \quad (X \neq Y), \qquad \left(T, T\right) = \left(P^{(T)}T, P^{(T)}T\right) + \left(P^{(S)}T, P^{(S)}T\right) + \left(P^{(q)}T, P^{(q)}T\right). \tag{A19}$$